Practical 1

Introduction to Wireshark

Learning Objectives

Upon completion of this practical, you will be able to:

- Configure a peer to peer network
- Test network connectivity
- Capture packets with Wireshark

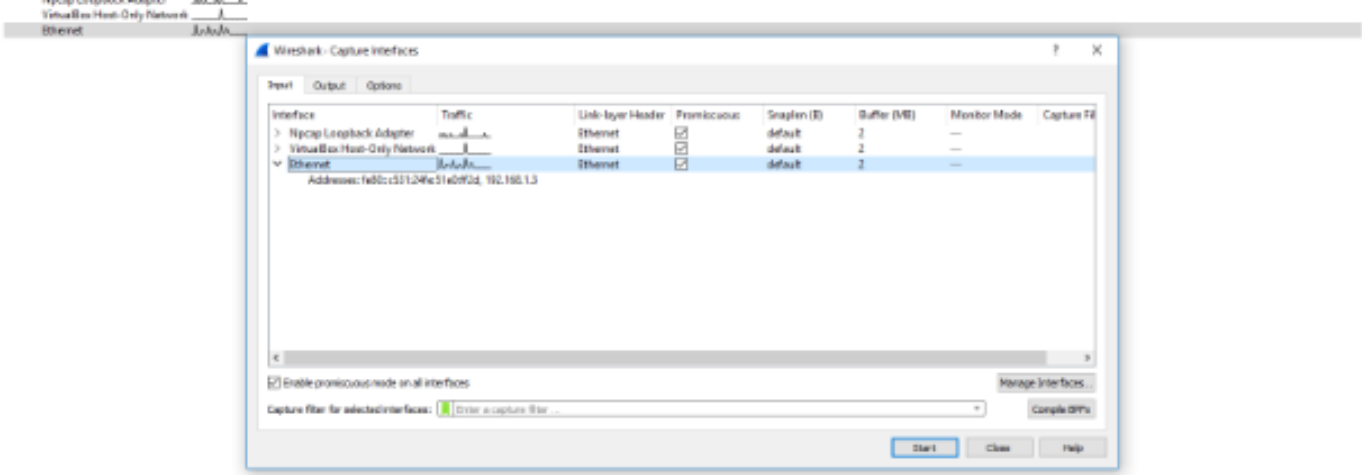
Background

A packet sniffer (also known as a network analyser or protocol analyser) is a computer software tool that can intercept and log data traffic passing over a data network. It can capture data or packets as the data streams travel back and forth over the network. Wireshark is one of free and open-source packet sniffers used for network troubleshooting, analysis, software and communications protocol development and education. If you are more interested you can visit the website "https://www.wireshark.org/" which provides detailed information about Wireshark.

In this unit, you start familiarising with Wireshark and use it to appreciate the entire process of data encapsulation. You will be able to use Wireshark to capture network traffic to analyse incoming and outgoing traffic, which is a critical part of network management activities. With this unit, you will be required to use Wireshark to perform numerous lab activities throughout the semester. You will be required to download it and install it on your device.

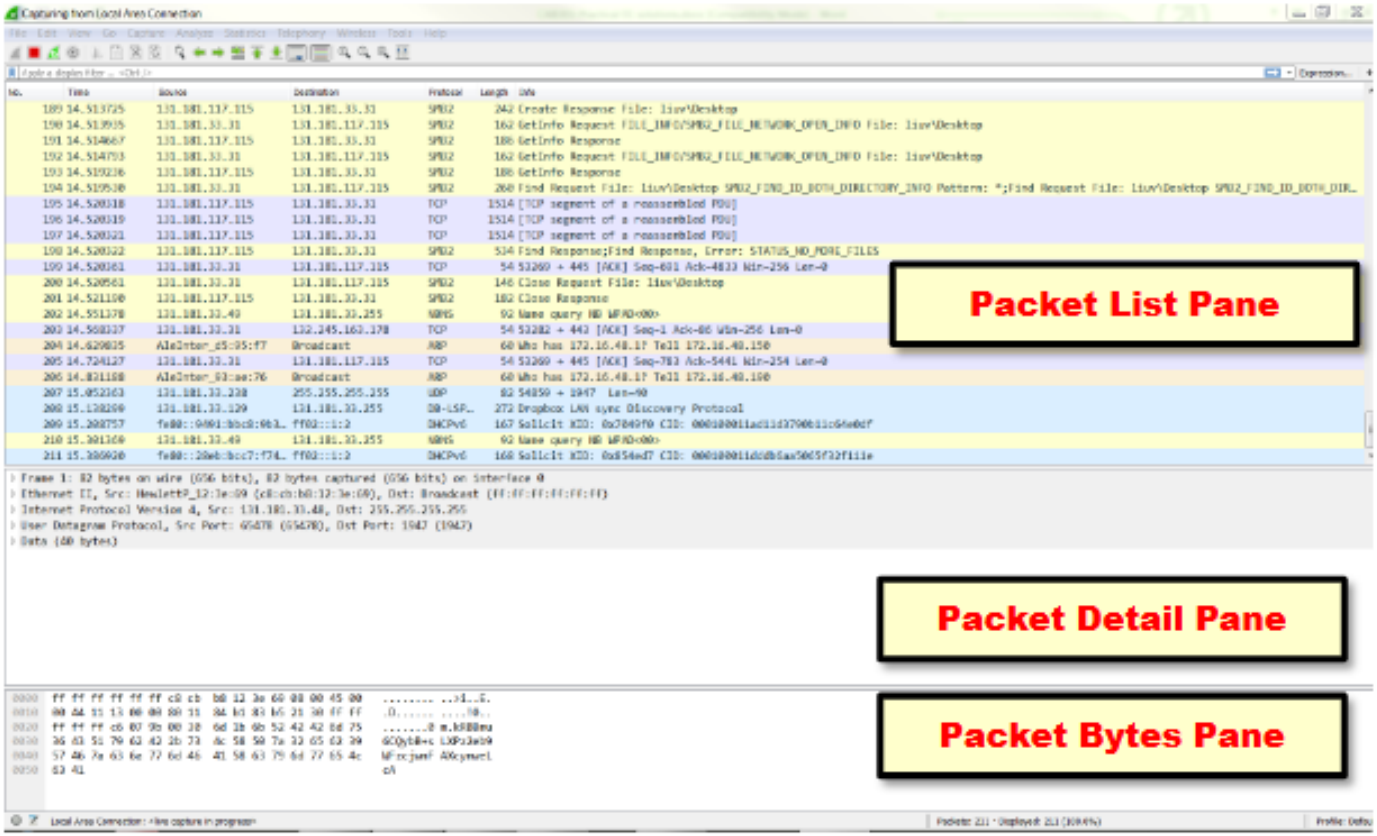
Lab Activities

- **Lab Activity 1**: Capture traffic between two PCs in a virtual lab environment.
- **Lab Activity 2**: Capture traffic on the QUT or your home network.



To start data capture, select "**Ethernet**" from the list of the Capture Interfaces to capture the network traffic occurring in the peer to peer local area network.

The main display window of Wireshark has three panes as shown in the following figure:



Practical 2

Introduction to Cisco Packet Tracer

Learning Objectives

Upon completion of this practical, you will be able to:

- Correctly identify cables for use in a network.
- Establish three different types of network topologies using Cisco Packet Tracer (PT).
- Verify basic connectivity on each network using PT.
- Understand how to use PT to build basic network topologies.

Background

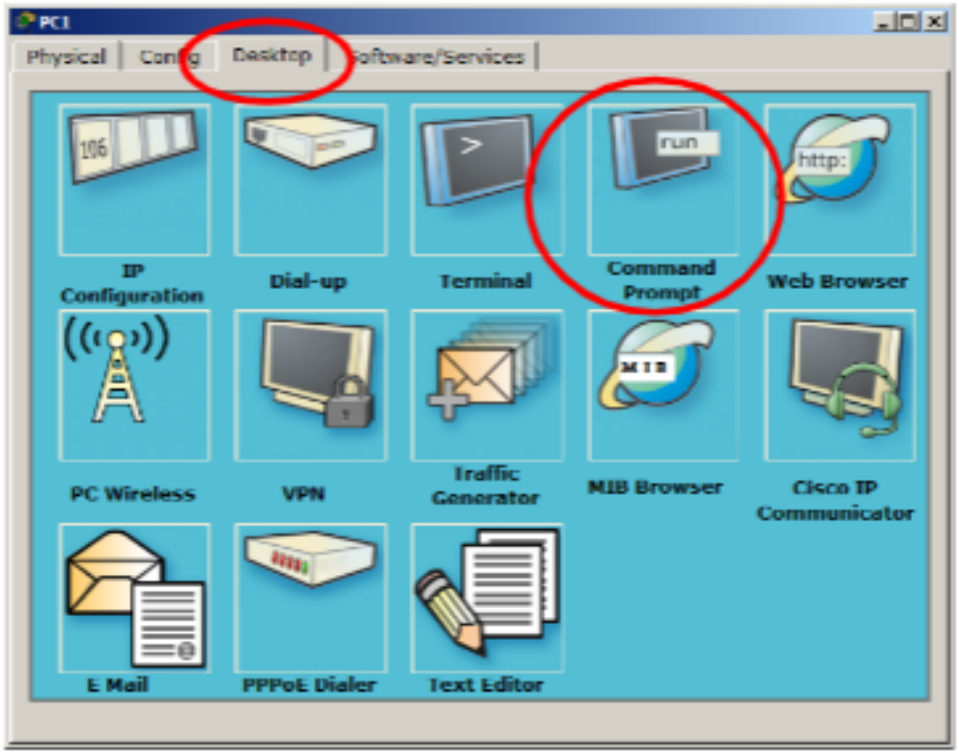
Cisco Packet Tracer (PT) is a software network simulator that allows you to virtually create, configure, and troubleshoot networking scenarios using either Cisco's Internetwork Operating System (IOS) command-line user interface (CLI) or graphical user interface (GUI) modes. PT allows you to observe and better understand how data (packets) travel and data is processed across a network, as well as enabling you to configure routers and switches using Cisco's IOS CLI.

This program is a simulator with some functional limitations; however, it is adequate for your learning in this unit. We will be using PT for numerous lab activities and using it to assess your practical skills. You will need to register to the Cisco Networking Academy to enable you to download Packet Tracer for free.

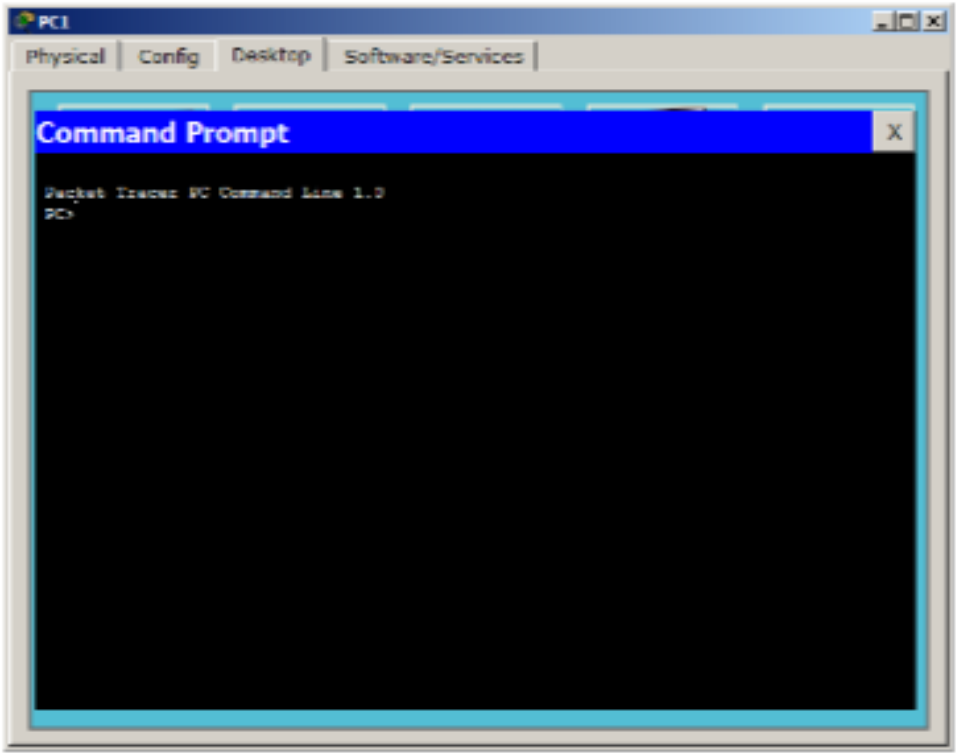
Lab Activities

In this practical, you are asked to use PT to set up three different network topologies and perform some basic configurations and then observe the network behaviour. In Lab Activities 2 and 3, you will be given an advanced scenario to challenge your learning.

- **Lab Activity 1**: Build a peer-to-peer network and then observe the network behaviour.
- **Lab Activity 2**: Build a switched network and then observe the network behaviour.
- **Lab Activity 3**: Connect two subnets using a router.



2. A command prompt window will appear.



Lab Activity 3 – Connect Two Subnets using a Router

Topology Diagram 3

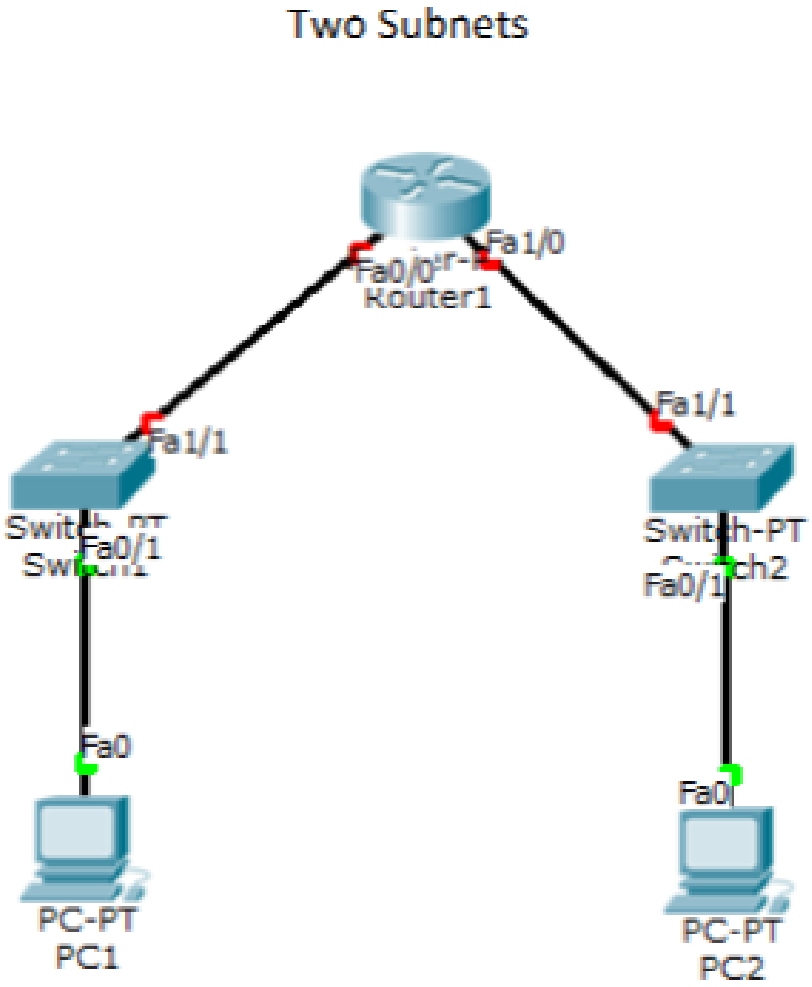


Figure 3

Practical 3

IPv4 Addressing

Learning Objectives

Upon completion of this practical, you will be able to:

- Identify the network information from a given IPv4 address and a subnet mask
- Calculate the subnet information from a given IPv4 address and a subnet mask
- Implement a subnetting scheme

- **Lab Activity 1**: Identify the network information from a given IP address and subnet mask
- **Lab Activity 2**: Subnetting Basics
- **Lab Activity 3**: Implement a subnetting scheme with Packet Tracer (PT)

	Network-ID			**Subnet-ID** & Host-ID
IP Address	11000001	00000010	00000001	**0**0001001
Subnet Mask	11111111	11111111	11111111	**1**0000000
Network Address	11000001	00000010	00000001	**0**0000000
First Host IP address	11000001	00000010	00000001	**0**0000001
Second Host IP address	11000001	00000010	00000001	**0**0000010
Third Host IP address	11000001	00000010	00000001	**0**0000011
• • •	• • •	• • •	• • •	• • •
Last Host IP address	11000001	00000010	00000001	**0**1111110
Broadcast Address	11000001	00000010	00000001	**0**1111111

TOPOLOGY DIAGRAM

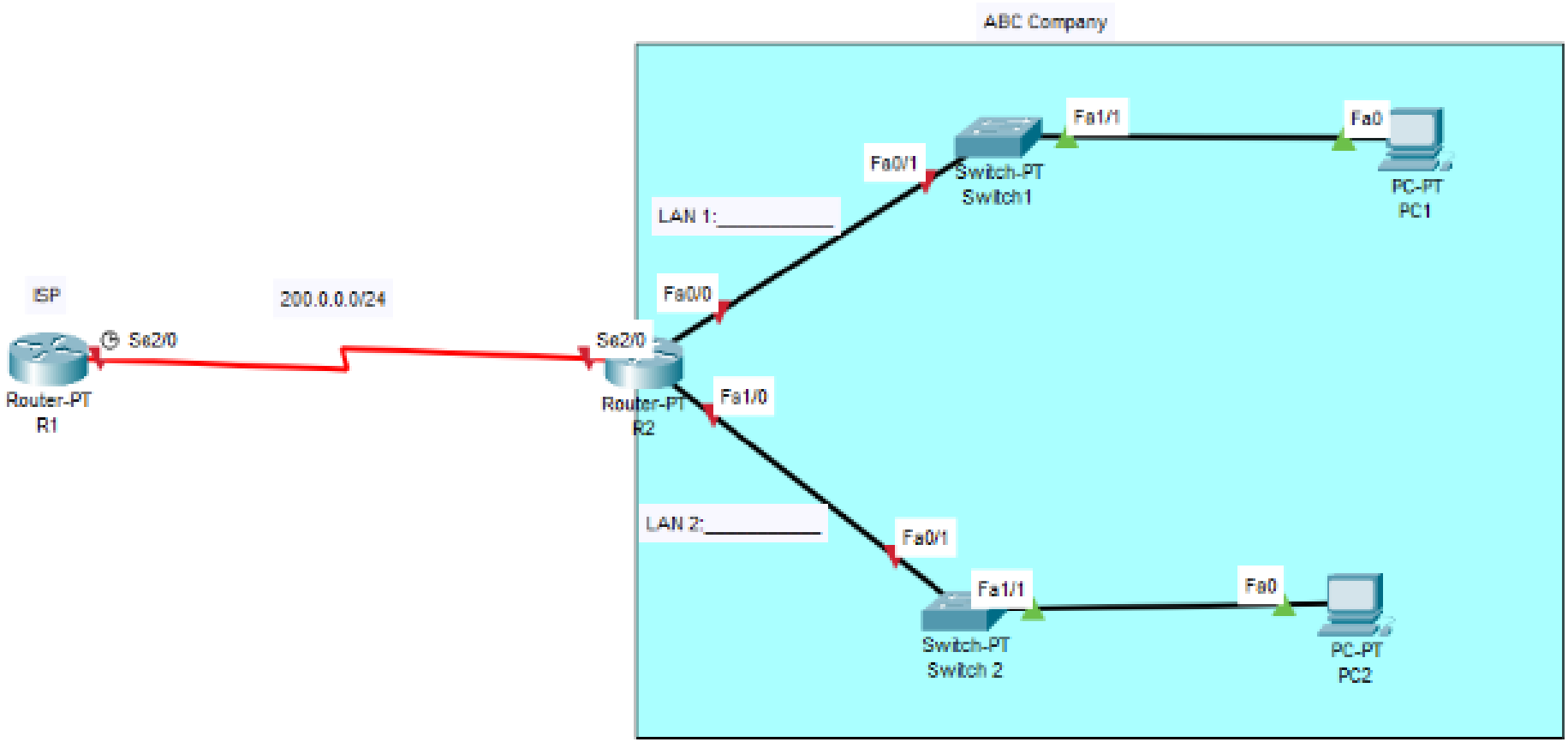


Figure 1 Network Topology